SECTION A

1. (a) A simply supported beam is subject to a uniformly distributed load q along CD as shown in Figure 1(a).

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**
   (ii) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tensile** side. **(5 marks)**

   (b) A three-hinge frame is subject to a pair of moments m. As illustrated in Figure 1(b), one clockwise moment is applied on the left of pinned joint E, while the other counter-clockwise moment is applied on the right of pinned joint E.

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**
   (ii) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tensile** side. **(5 marks)**

2. A simply supported beam with overhang is subject to a point load F on C and a uniformly distributed load q along BD as shown in Figure 2(a). The cross section of the beam is made up of T-section, see Figure 2(b). The load F is inclined at an angle of 45 deg.

   (a) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tensile** side. **(4 marks)**

   (b) Draw the axial force diagram (AFD) with the sign labels and pertinent values. Note that positive axial force is along the **tensile** direction. **(4 marks)**

   (c) Find the maximum tensile stress, maximum compressive stress and their corresponding positions in the T-beam. **(8 marks)**

   (d) Check the strength condition according to the maximum shear stress (Tresca) criteria and find the position where the maximum shear occurs. The allowable shear stress $[\tau]$ = 185 MPa. **(4 marks)**

3. Two trusses, truss 1 and truss 2, are subject to point loads P_1 and P_2, as shown in Figure 3 (a) and (b) respectively.

   (a) Calculate the axial forces of all the bars in truss 1 and truss 2. The axial forces should be expressed in terms of P_1 or P_2. **(10 marks)**

   (b) Use Euler's buckling theory to calculate the critical buckling loads $P_{1,cr}$ and $P_{2,cr}$ of the two trusses, respectively. Also point out which bars will buckle for the two trusses. For each of the members, EI = 500 kN m^2. **(10 marks)**

   Hint for 3(b): For buckling analysis, each truss bar can be treated as a beam with both ends pinned. According to Euler's formula, the critical load for this type of beam with span L is $P_{cr} = \dfrac{\pi^2 EI}{L^2}$.

FIGURES

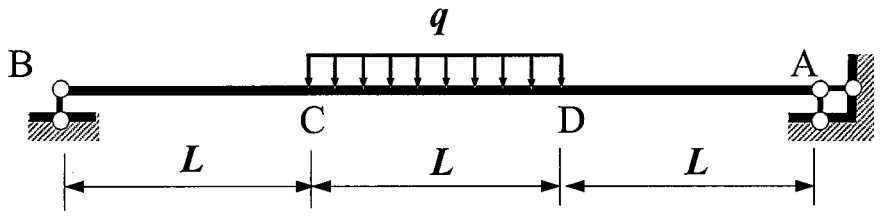


Figure 1(a) Beam

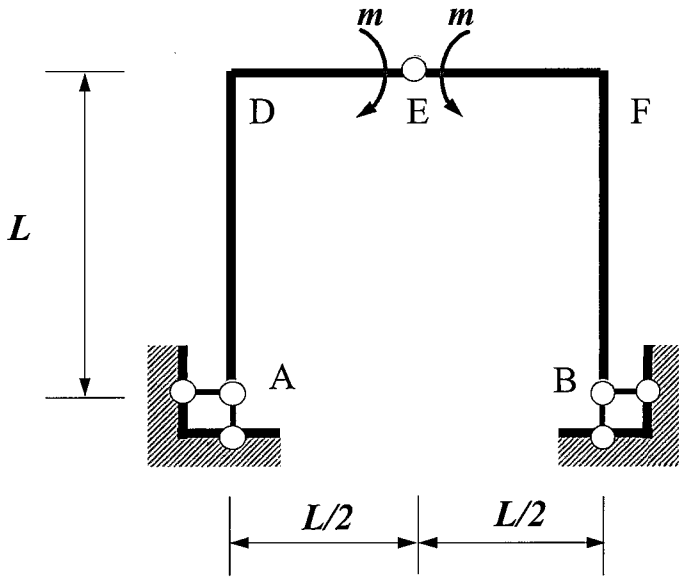


Figure 1(b) Three-Hinge Frame

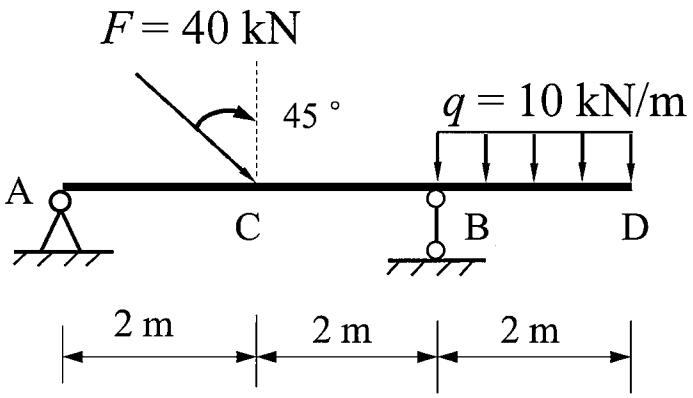


Figure 2(a) Beam with Overhang

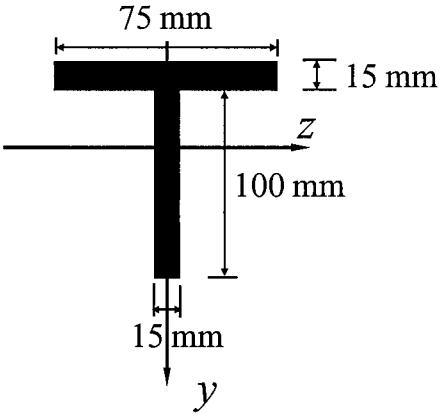


Figure 2(b) T-Section

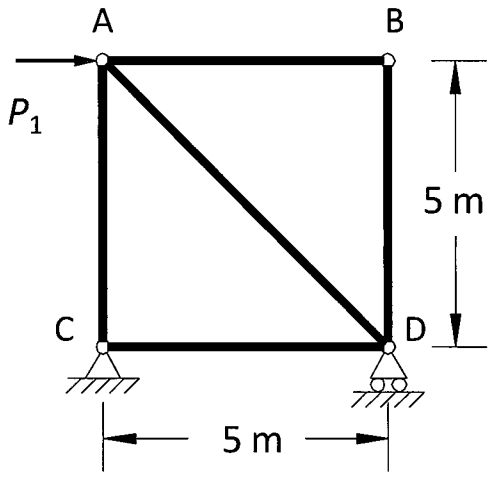


Figure 3(a) Truss 1

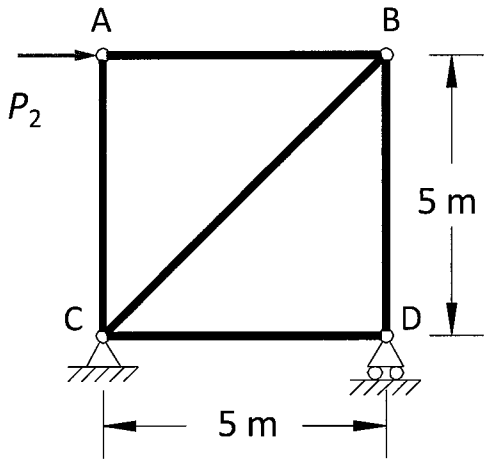


Figure 3(b) Truss 2

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